

Name: _____ Mark: _____ /67

G10 Science - Class 4 Homework

1. What are the most obvious differences between prokaryotes and eukaryotes? **[2 marks]**

2. What are the main differences between an animal cell and plant cell? **[3 marks]**

3. Plant cells can make their own food through photosynthesis. Why do plant cells still have mitochondria? **[3 marks]**

4. Why do cells divide rather than get bigger as an organism grows? **[2 marks]**

5. Which stage of the cell cycle corresponds to each of the descriptions below? **[6 marks]**

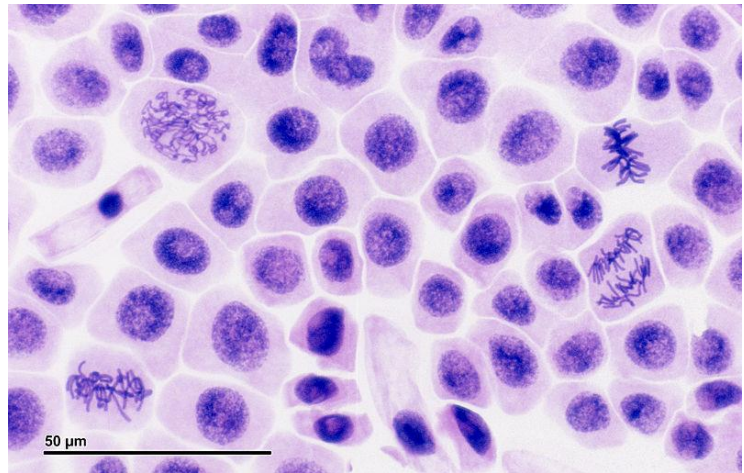
A new cell wall begins to form	
The membrane of the nucleus dissolves	
Daughter chromosomes begin to separate	
The cell has completely pinched together along its centre to form two separate cells	
The cell grows and copies its DNA	
The chromosomes are aligned in the middle of the cell	

6. Label the following diagrams with the correct phase in mitosis. **[4 marks]**



7. Using the follow slide, identify a cell that is undergoing: **[5 marks]**

- A) Interphase
- B) Prophase
- C) Metaphase
- D) Anaphase
- E) Telophase



8. For the slide in Question 7, is this sample most likely derived from an animal or plant? Explain your reasoning. **[2 marks]**
9. Some antibiotics interfere with a bacterial cell's ability to copy DNA.
- a) How would this type of antibiotic be able to stop a bacterial infection? **[2 marks]**
 - b) These antibiotics do not have any effect on the DNA replication of human cells. What makes bacterial cells different from human cells? **[2 marks]**

10. How are cancer cells different from normal cells? **[3 marks]**

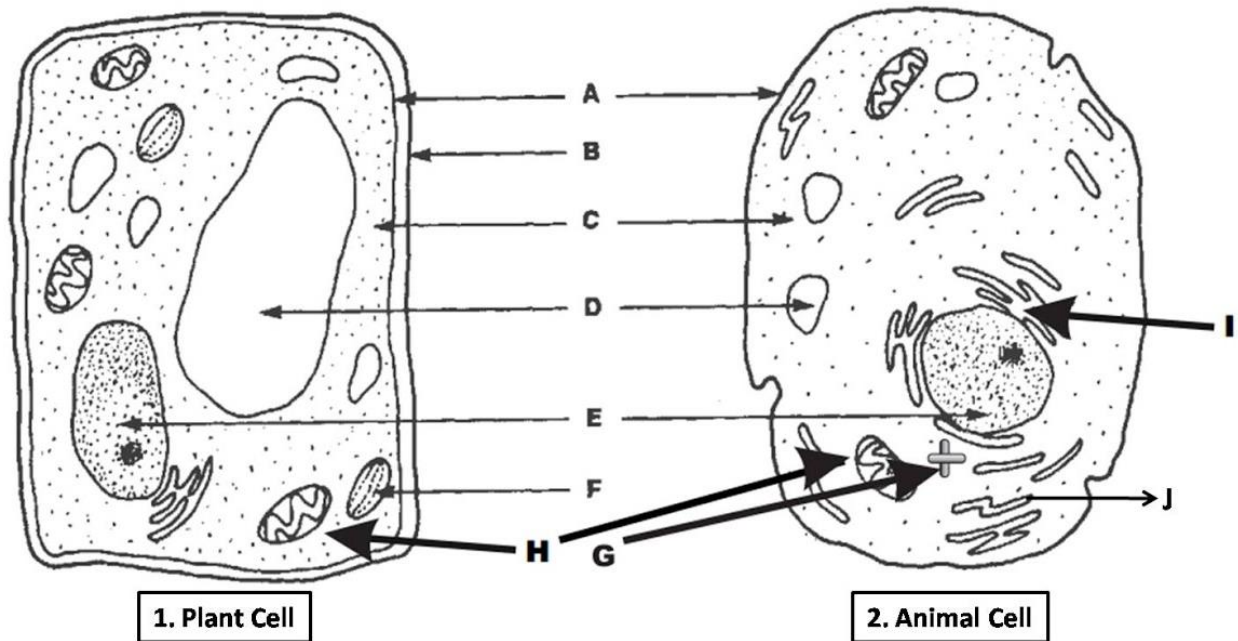
11. Why might it be easy to overlook cancer in its early stages? What preventative measures can be taken to prevent late-stage cancers? **[3 marks]**

12. What is the difference between an embryonic stem cell and an adult stem cell? **[2 marks]**

13. Complete the table by filling in the appropriate information: **[8 marks]**

Tissue Type	Structure	Function
	Made up of cells that are able to contract	
		Transmitting information around the body
Connective		
		Protection and reduction of water loss

14. Label the following diagram and describe the function of each organelle. [20 marks]



	Organelle	Function
A		
B		
C		
D		
E		
F		
G		
H		
I		
J		